Paper. F. N. Nagoor Pitchai, E. J. Tanner, N. Khetan *et al.*, "Engineered deletions of HIV replicate conditionally to reduce disease in nonhuman primates," *Science* **385**, eadn5866 (2024).

1. What they built, and the model behind the figure

They isolated an HIV defective interfering particle from a non-dilutive continuous culture reactor: a ∼2.5 kb deletion in *pol–vpr* (RNase H and integrase removed, reverse transcriptase truncated). It had $R_0 < 1$, making up 60% of packaged extracellular RNA but only 30% of integrated provirus in the next round (Fig. 1D). Reintroducing the central polypurine tract and ablating *tat/rev/vpu/env* gave TIP-2, with $R_0^{\mathrm{TIP}} = 12 \pm 0.5$ by three-colour coculture (Fig. 2F). An SIV analogue, one IV injection into infant rhesus macaques challenged 24 h later with SHIV-SF162P3, held the set point near 10^4 copies/mL in five of six animals against $\sim 10^8$ in controls over 30 weeks.

The figure they lean on hardest: Figure 4D, summarised in 4E. A reduced inoculum, an attenuated variant and innate control are excluded by experiment (Fig. 3C, full-length sequencing, fig. S14); the model supplies the positive mechanism. The basic model fits the controls but no treated animal under any parameter combination, the discriminating statistic being the peak-to-set-point ratio in $\log_{10}$, while the TIP extension fits both. That fit licenses the human extrapolation of Fig. 5G.

Basic model (Perelson *et al.* 1996; Nowak & May 2000; refs 24, 25), with target cells T, infected cells I and free virus V: $\dot{T} = \lambda - dT - \beta VT$, $\dot{I} = \beta VT - \delta I$, $\dot{V} = N\delta I - cV$.

TIP extension (Weinberger, Schaffer & Arkin 2003, ref 7). Six states: targets T, HIV-infected I_H, TIP-only I_T, dually infected I_D, free virus V_H, free TIP V_T:

$$\begin{aligned} \dot{T} &= \lambda - dT - \beta_H V_H T - \beta_T V_T T, & \dot{I}_D &= \beta_T V_T I_H + \beta_H V_H I_T - \delta_D I_D, \\ \dot{I}_H &= \beta_H V_H T - \delta I_H - \beta_T V_T I_H, & \dot{V}_H &= N\delta I_H + (1-D)\, N\delta_D I_D - cV_H, \\ \dot{I}_T &= \beta_T V_T T - d_T I_T - \beta_H V_H I_T, & \dot{V}_T &= \Pi\, N\delta_D I_D - cV_T. \end{aligned}$$

Two composite parameters carry the design: D, the inhibition of wild-type output from a dually infected cell, and Π, the packaging advantage (P in ref 7). Conditional replication is the absent production term in I_T.

2. What I would build

A TIP maintained by division of its host cell rather than by packaging.

The starting observation. The TIP mobilises only from dually infected cells, so its reproduction scales with V_H. In suppressing the virus it starves its own source: the system runs to coexistence, not clearance, and there is a floor viral load below which the TIP cannot push without extinguishing itself. The paper calls non-clearance a limitation of conditional replication; it is a design variable.

The obvious knobs do not fix it. Fig. 2H shows deleting Ψ blunts interference by 31%, so D is largely downstream of packaging competition and cannot be raised while Π is lowered; and raising either lowers V_H, starving the TIP further.

The exit is a maintenance route that does not pass through Gag. Memory CD4$^+$ T cells self-renew and an integrated provirus is copied with the genome: this, not virion production, is how the latent HIV reservoir persists for decades. A TIP partitioned into that compartment would be replicated by the cell while producing nothing without HIV, leaving conditional replication and its safety argument untouched. Superinfection then supplies both halves at once, Tat to reverse the TIP's latency and Gag to package it, the paper's own account of lentiviral latency (refs 62–64).

The device: transduce activated CD4$^+$ cells, as the paper already does, and configure the vector so that transduced cells partition into the central-memory pool rather than the short-lived effector pool. Delivery and superinfection stay in activated cells, which are permissive; only maintenance moves.

3. What I would have to compute

A TIP-only cell is superinfected before dying with probability $p = \beta_H V_H/(\beta_H V_H + d_T)$. With $A = \Pi N \beta_T T^*/c$ the TIP's reproductive capacity conditional on superinfection, $R_0^{\mathrm{TIP}} = pA$, so the floor satisfies $p^* = 1/A$ and $V_H^* = d_T/[\beta_H(A-1)]$.

First, a check that needs no simulation. The threshold superinfection probability is $1/A$, independent of d_T. With $A = 12$ that is 8.3%, and the paper's Discussion cites 5 to 10% of infected cells carrying multiple proviruses at chronic set point. The published TIP would then sit at $R_0^{\mathrm{TIP}} \approx 1$ in humans rather than safely above it, which is the margin Fig. 5G needs. Both numbers are in the paper. Converting that frequency into p is the first task, since it is conditioned on infection and cumulative over a cell's life. $A = 12$ is also a ceiling: the Fig. 2F assay starts from producer cells already transduced and then infected, so superinfection is set up rather than sampled.

Second, what the design turns on. Give the TIP-only compartment the proliferation term the paper's model omits, $\rho(1-\varepsilon)I_T$, with ρ the self-renewal rate of the memory pool and ε the fitness cost of carrying the TIP. The compartment is then maintained with no packaging input whenever $\rho(1-\varepsilon) > d_T + \beta_H V_H$, so the floor is removed rather than lowered, and the design reduces to a tolerance $\varepsilon_{\mathrm{crit}} = 1 - (d_T + \beta_H V_{\mathrm{target}})/\rho$ at $V_{\mathrm{target}} = 10^3$ copies/mL. $\varepsilon_{\mathrm{crit}}$ is computable from published memory CD4$^+$ self-renewal and loss rates. What it must be compared against is not yet measured: fig. S16E and fig. S6B–C both compare TIP-transduced with control cells *in the presence of HIV*, so the intrinsic cost and the protection from HIV killing enter the same number with opposite signs, so the durability Fig. 5G assumes rests on an unbounded parameter. Establishing $\varepsilon_{\mathrm{crit}}$, and what would bound ε, is the project.